

Amine Hamlouchi

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EDUCATION

University of Louisville, J.B. Speed School of Engineering

Bachelor of Arts in Computer Science; Minor in Arabic; GPA: 3.5/4.0

Louisville, KY

Expected May 2028

HackKentucky 2026 (KY Combinator, Louisville KY) | 3rd overall and the Prologue Stories sponsor bounty, for InterLogueSep.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL, Bash, HTML/CSS

AI/ML & Data: RAG (retrieval-augmented generation), LLM orchestration, NLP / information extraction, ETL pipelines, pandas, NumPy, scikit-learn

Web & Mobile: React, React Native, Next.js, Node.js/Express, FastAPI, REST APIs, PostgreSQL/PostGIS, Supabase

Infrastructure: Docker/Podman, GCP Cloud Run, Vercel, Pulumi (IaC), GitHub Actions (CI/CD), Firebase Auth, Stripe, Git, Linux

EXPERIENCE

Product Engineering Intern

Aug. 2026 – Present

Kamel Ride

Remote

- Ship features and fixes in a production monorepo (Node/Express with PostgreSQL + PostGIS, React/TypeScript web, React Native mobile) deployed to GCP Cloud Run via GitHub Actions with Pulumi-managed infrastructure.
- Diagnosed and fixed an upload defect where the client accepted files the API rejected (10MB UI vs. 5MB server cap) and silently stored blank HEIC images; shipped client-side validation and in-browser HEIC-to-JPEG conversion.
- Own the React Native messaging feature; traced a broken media rollout to un-applied Pulumi state and reported the root cause and remaining cutover steps to the engineering lead.

IT Analyst Intern

May 2026 – Present

Parker Hannifin

Jeffersonville, IN

- Troubleshoot endpoint hardware, network, account access, and enterprise applications for on-site manufacturing teams, isolating root causes and documenting repeatable fixes.
- Write PowerShell, Batch, Bash, and SQL scripts to automate routine diagnostics, endpoint configuration, and data lookups.

Technical Lead, Digital Operations

2024 – Present

Alnur Mosque Islamic Center

Louisville, KY

- Designed and shipped the organization's 11-page public website (semantic HTML/CSS, no framework) on GitHub Pages: programs, events, online booking, donations, and bilingual English/Spanish resources.
- Built a zero-developer publishing pipeline (Python/Pillow + GitHub Actions): volunteers drop photos in a folder and CI deduplicates by SHA-256, corrects EXIF orientation, re-encodes, and regenerates the gallery manifest on push.

PROJECTS

InterLogue (interloguemcp.com) | MCP, Claude, TypeScript/Node, ElevenLabs, Twilio

2026

- Founded and built an MCP tool inside Claude that is briefed like a reporter, places a real phone call, runs a live interview, and returns a publishable case study with every quote cited to a timestamp in the transcript; took 3rd overall and won the Prologue bounty at HackKentucky 2026.
- Engineered consent as a hard gate rather than a disclaimer: the agent states it is an AI and asks permission to record, and the transcript is destroyed without an explicit yes, which caught and discarded a real call when the opening script regressed.

Argument Knowledge Base (RAG) | Python, NLP, LLM Orchestration, REST APIs

2026

- Built an end-to-end NLP pipeline turning 32 multi-hour audio debates into a citation-verified RAG knowledge base of 731 canonical arguments and 893 paired rebuttals (1,100+ citations), with two-level fact verification validating 712 citations against 4 primary-source APIs and dropping unverifiable claims; orchestrated 8–12 parallel LLM sub-agents across an idempotent, checkpointed pipeline over ~1,000 records.

School Operations Platform | TypeScript, Next.js, Supabase (Postgres), Vercel

2026

- Designed, built, and launched a production bilingual (English/Arabic, full RTL) school-operations platform serving 80 active users across admin, teacher, family, and student roles, replacing a year of manual spreadsheets with one PostgreSQL system of record.
- Built its TypeScript serverless backend (106 API routes, 15 Postgres migrations) covering grading, attendance, scheduling, payroll, and tuition, with role-based access enforced by PBKDF2 hashing and Postgres row-level security and 51 regression tests.

Rumi: Roommate-Matching Platform (findyourrumi.com) | TypeScript, Next.js, PostgreSQL, Vercel

2026

- Founded and shipped a privacy-first roommate-matching platform end to end, with a compliance-gated release architecture and a double opt-in waitlist (hashed single-use tokens, 18+ validation).